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Extra View

# Signalling molecules, growth regulators and cell cycle control in Drosophila

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**Key words:** Notch, Wingless, bantam, micro-RNA, E2F, dMyc

**During the development of a given organ or tissue within a multicellular organism, growth and patterning are controlled in a coordinated manner by the activity of a discrete number of signalling molecules and their corresponding pathways to give rise to a well-formed structure with a particular size, shape and pattern. Understanding how cells of different tissues or organs translate the activity of these pathways into an activation or repression of the cell cycle machinery in a context-dependent manner is at present one of the most intriguing questions in developmental and cancer biology. In this article we review the different roles of the Notch and Wingless signalling molecules in the regulation of cell cycle progression in the developing eye and wing imaginal discs of Drosophila and propose that, depending on how growth regulators are controlled in a context-dependent manner by the activity of these pathways, signalling molecules might have tumour suppressor or oncogene activity.**

## The DV Organizing Centre in the Drosophila Wing Primordium

The Drosophila wing primordium arises as a group of 30–40 cells in the embryonic ectoderm that proliferates for five days to reach a final size of around 50,000 cells.[1,2] During metamorphosis, the wing disc gives rise to the adult wing and part of the thoracic body wall. The adult wing has a species-specific size, shape and pattern. Growth of the wing primordium is intrinsically controlled by the activity of a number of signalling pathways that mediate short-range as well as long-range interactions among proliferating wing cells. Signalling molecules that mediate long-range effects in growth are expressed along the so-called compartment boundaries, which behave as growth organizing centres in the developing primordia. When the wing disc has grown into a few hundred cells, it subdivides into a dorsal (D) and a ventral (V) cell population or compartment by the activity of the LIM-Homedomain transcription factor Apterous in D cells (Fig. 1A).[3,4] Cell interactions between D and V cells lead to activation of the Notch signalling pathway at the compartment boundary.[5-7] Notch activity induces the expression of the Wingless signalling molecule, the founding member of the Wnt family of secreted proteins (Fig. 2B).[5,6] The DV boundary has a role in organizing growth of the whole wing primordium.[3] In the absence of Notch activity at the DV boundary, the wing primordium does not grow, and ectopic activation of Notch in non-boundary cells induces non-autonomous overgrowths.[8] Wg is expressed at the DV boundary, it symmetrically diffuses at both sides of the boundary and acts at a long-range distance to activate target gene expression.[9,10] Wg has been proposed to mediate the organizing activity of the DV boundary,[6] although its role has become controversial.[11-13] Thus, we are still

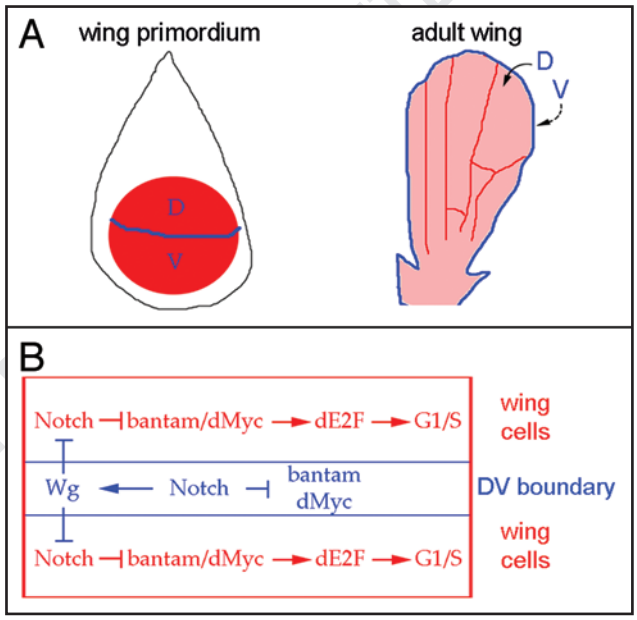


Figure 1. Regulation of $G_1$-S transition by Notch and Wg in the Drosophila wing. (A) The wing primordium (left) subdivides into dorsal (D) and ventral (V) compartments that will give rise to the D and V surfaces of the adult wing (right). The region that gives rise to the adult wing is labelled in red and the DV boundary in blue. (B) Activation of Notch along the DV boundary leads to the expression of the secreted molecule Wingless (Wg). The activity of Notch and Wg signalling pathways spatially regulates the activity of *bantam* micro-RNA and the expression levels of the *dMyc* proto-oncogene. *bantam* and dMyc mediate regulation of the E2F transcription factor and the transition of $G_1$ to S phase.

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far from understanding which factors mediate the organizing activity of the DV boundary.

## Wg and Notch have Opposite Effects on $G_1$-S Progression

During larval development, wing disc cells proliferate at random. Late in development, cells at the DV boundary are characterized by cell cycle arrest and define the so-called Zone of Non-proliferating Cells (ZNC).[14] Boundary cells are arrested in $G_1$,[15] whilst the rest of the cells in the wing primordium continue to proliferate actively.[16,17] The pattern of cell cycle proliferation at these stages correlates with the pattern of activity of the mutually exclusive Notch and Wg signalling pathways. In boundary cells, which are cell cycle-arrested, Notch is active.[5,6] In non-boundary cells, which are actively proliferating, Wg pathway is active[9,10] and the response to Notch is reduced by the activity of Wg.[18] Thus, cell cycle progression in the late wing primordium might be governed by the opposite effects of Notch and Wg (Fig. 1B). Consistent with that, the $G_1$ cell cycle arrest that takes place at the DV boundary is defined by the activity of Notch, as boundary cells unable to respond to Notch enter into S phase and ectopic activation of Notch in non-boundary cells imposes a cell-autonomous cell cycle arrest in $G_1$.[19] In non-boundary cells, the situation is more complex. Wg acts by repressing Notch response, thus alleviating the block in $G_1$. Non-boundary cells unable to respond to Wg activate the Notch pathway and get arrested in $G_1$. When these are also unable to respond to Notch, then they cycle through $G_1$. E2F transcription factors act as universal $G_1$-S master regulators and regulate the expression of a number of genes required for $G_1$-S transition, including Cyclin E, as well as S phase, and their activity is inhibited by the Retinoblastoma (Rb) proteins like Drosophila Rbf (Rb Familiy Protein, reviewed in ref. 20). The Rb pathway is a key regulator of the $G_1$-S transition and Drosophila Rbf is required for $G_1$ arrest in the ZNC.[21] The opposing activities of Notch and Wg on $G_1$-S transition are the sole consequence of regulation of E2F activity.[19]

Notch imposes a $G_1$ cell cycle arrest in DV boundary cells and Wg activity is required in wing blade cells to repress Notch signalling and allow cells to enter the S phase, and they do so through regulation of the E2F transcription factor. The fact that cells unable to respond to Notch and Wg do proliferate indicates that Wg has a permissive role in cell cycle progression in the Drosophila wing and implies that unknown Notch-regulated signalling molecules might exert the organizing activity of the DV boundary.

## Effects of Wg and Notch in Cell Cycle Proliferation in other Developmental Contexts

Although Wingless has a permissive function in alleviating the $G_1$ cell cycle arrest imposed by Notch activity in the wing pouch, its activity is required to promote cell proliferation in the proximal part of the wing, the so called wing hinge,[22] and to impose a cell cycle arrest in the precursor cells of the sensory organs along the DV boundary.[15] In this latter context, Wg imposes a cell cycle arrest in $G_2$ by inhibiting the expression of *Dcdc25/string*, the universal eukaryotic regulator of the $G_2$/M transition.[23] In the wing hinge, reduction of Wg activity leads to under-proliferation of this region and loss of hinge structures.[22] Conversely, overexpression of Wg or activation of the Wg signal transduction pathway leads to local overgrowth of the proximal hinge. The effects of Wg in the proximal hinge suggest a rather direct mitogenic effect in this region of the disc, although the cell cycle elements regulated by Wg in this region remain to be elucidated.

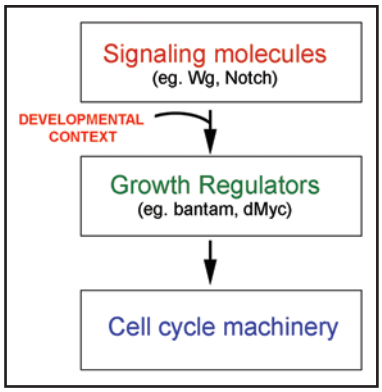


Figure 2. Context-dependent regulation of the cell cycle machinery. Activity of signalling molecules is translated, in a context dependent manner, into the activation or repression of the cell cycle machinery. The context might be conferred by the ability of these pathways to regulate, either positively or negatively, the expression of growth regulators such as *bantam* or *dMyc*.

Similarly to Wingless, the effects of Notch on cell cycle proliferation depend on the developmental context. While Notch imposes a cell cycle arrest in $G_1$ by inhibiting the activity of the E2F transcription factor in the wing, Notch acts as a positive regulator of E2F in the developing eye primordium of Drosophila.[24-26] In the Drosophila eye, retinal development exemplifies patterned cell proliferation. A wave of differentiation, known as the morphogenetic furrow (MF), spreads progressively across the eye primordium (reviewed in refs. 27 and 28). Anterior to the furrow, cells proliferate randomly, and there is no overt differentiation. Growth of this tissue is organized by the boundary between dorsal and ventral compartments. The organizing activity of this boundary is executed by Notch. Ectopic activation of the Notch pathway in this tissue leads to tumour formation and this is due to Rbf depletion and E2F hyperactivation.[26] Posterior to the morphogenetic furrow, ommatidial clusters start to form. As cells enter the MF, they get arrested in $G_1$. A few hours later, as they emerge from the posterior of the furrow, cells can subdivide into two populations: precluster cells that initiate ommatidial recruitment and undifferentiated surrounding cells. Those cells that have not joined a precluster undergo one final and simultaneous round of mitosis, needed to produce enough cells to complete development. This final round of mitosis is called the second mitotic wave (SMW) (reviewed in ref. 27). Notch is required in this context to overcome the $G_1$ cell cycle arrest and trigger the initiation of the SMW. Notch exerts its function by positively regulating E2F activity and *cyclin A* expression, two effectors that regulate $G_1$-S progression.[24,25]

## Available Growth Regulators Might Translate the Activity of Signaling Molecules into Activation or Repression of the Cell Cycle Machinery

Here we have summarized the different outputs of Notch and Wg activities in the regulation of cell cycle progression in the developing eye and wing imaginal discs of Drosophila. Depending on the developmental context, they might act as oncogenes or tumour suppressor genes. In the wing primordium, regulation of E2F activity and $G_1$-S progression by the activities of Wg and Notch are mediated

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by two independent growth regulators: the *bantam* micro-RNA and the *dmyc* proto-oncogene (Fig. 1B).[19] The activity of *bantam* micro-RNA, which is involved in growth and cell survival in Drosophila tissues,[29] and the expression of the *dMyc* proto-oncogene, which regulates $G_1$-S progression in wing cells,[30] are spatially regulated by the activities of Wg and Notch, and most interestingly, execute the regulation of E2F activity by both signalling molecules. We would like to speculate that depending on the available growth regulators such as *bantam* and *dMyc*, Wg and Notch, as well as other signalling molecules, might exert a positive or negative regulation of the cell cycle machinery (Fig. 2). The growth regulators that mediate the activity of these signalling molecules in the eye remain to be elucidated. Finally, we would like to conclude, according to the summarized results, that signalling molecules should not be defined as tumour suppressor genes or oncogenes, since their activity depends absolutely on the developmental context.

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